Perceptron with Hebbian Learning - Report

Brum

March 19, 2026

1 Introduction

This report analyzes the performance of a Perceptron trained using the Hebb rule on three different datasets. The model was evaluated based on accuracy, training error evolution, and decision boundary visualization (for 2D datasets).

2 Dataset 1

2.1 Results

- **Training Accuracy (Last Epoch):** 96.43%
- **Test Accuracy:** 95.00%

2.2 Analysis

The model achieved high accuracy on both training and test sets. As observed in the decision boundary plots below, the data is almost linearly separable, allowing the Perceptron to find a hyperplane that correctly classifies most samples. The training error converged quickly (as seen in the error evolution graph).

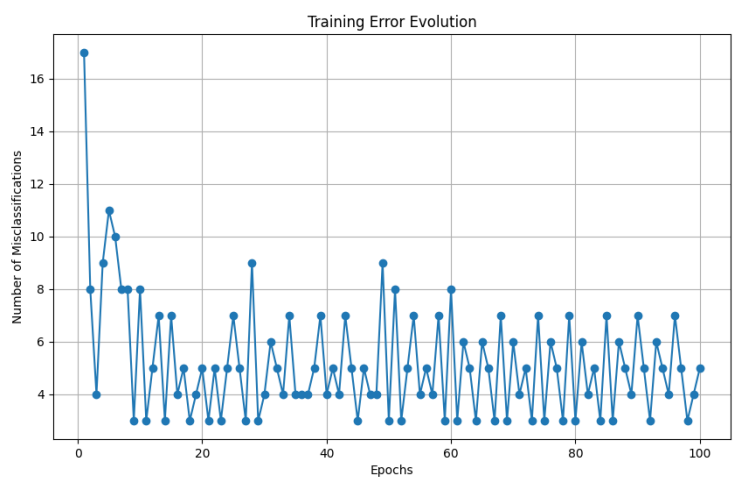
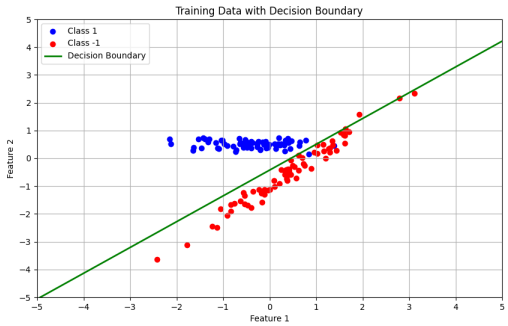
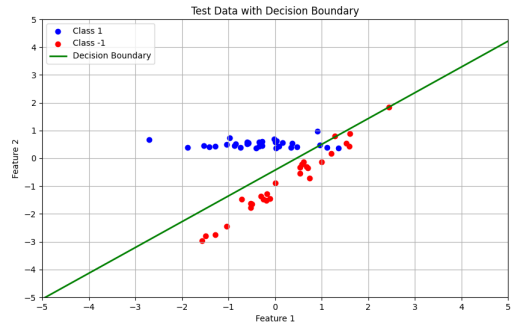


Figure 1: Dataset 1: Training Error Evolution



(a) Training Data



(b) Test Data

Figure 2: Dataset 1: Decision Boundary

3 Dataset 2

3.1 Results

- **Training Accuracy (Last Epoch):** 49.71%
- **Test Accuracy:** 68.00%

3.2 Analysis

The performance on Dataset 2 was poor, with accuracy hovering around random guessing (50-60%). This is expected due to the geometric distribution of the data.

The dataset consists of two classes arranged in a concentric manner: one class forms a smaller inner circle, while the other forms a surrounding ring (disk). This distribution is inherently **non-linearly separable**. Since the Perceptron is a linear classifier, it can only draw a straight line to separate the classes. It is impossible to separate a circle from a ring with a single straight line, resulting in the high classification error and the "cut-through" decision boundary seen in the plots.

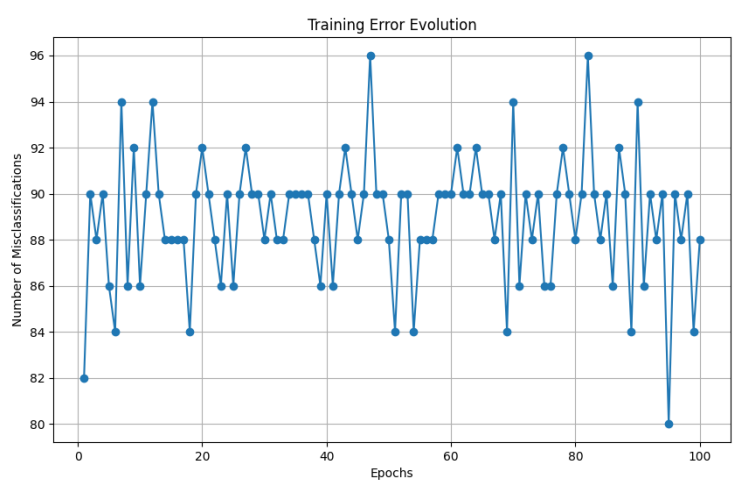
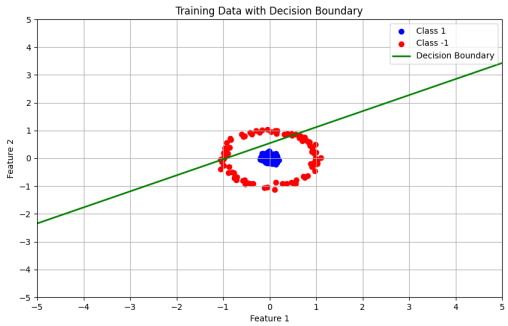
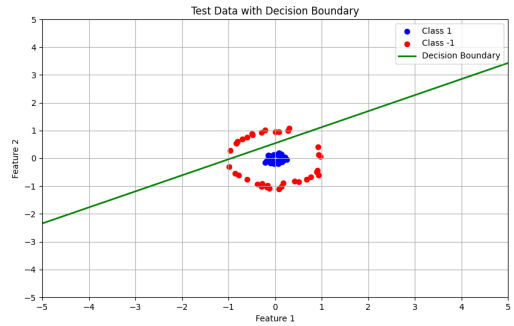


Figure 3: Dataset 2: Training Error Evolution



(a) Training Data



(b) Test Data

Figure 4: Dataset 2: Decision Boundary (Linear Attempt on Non-Linear Data)

4 Dataset 3

This dataset has 10 attributes, so decision boundaries cannot be visualized in 2D. We performed a grid search over Learning Rate (LR) and Epochs, repeating each configuration 20 times to calculate mean accuracy and standard deviation.

4.1 Results Summary

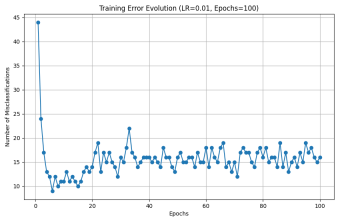
Learning Rate	Epochs	Mean Test Accuracy	Std Deviation
0.01	100	83.97%	2.40%
0.01	200	83.17%	1.77%
0.01	400	84.29%	2.18%
0.1	100	82.70%	2.70%
0.1	200	84.29%	3.51%
0.1	400	83.81%	2.91%
0.5	100	83.17%	3.11%
0.5	200	83.97%	2.50%
0.5	400	83.81%	2.33%

Table 1: Performance on Dataset 3 with varying hyperparameters (20 runs)

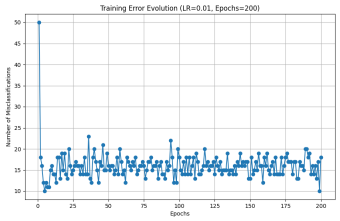
4.2 Analysis

- The highest mean test accuracy (84.29%) was achieved with $\eta = 0.1$ (200 epochs) and $\eta = 0.01$ (400 epochs).
- **High Variance:** All configurations show a standard deviation between 1.77% and 3.51%, indicating that the final accuracy depends somewhat on the random initialization of weights.
- **Stability:** Increasing epochs to 400 yielded consistent results across different learning rates, suggesting convergence.

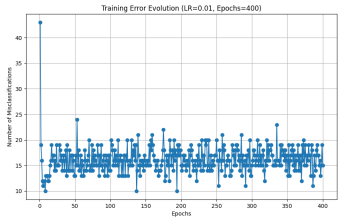
4.3 Training Error Plots



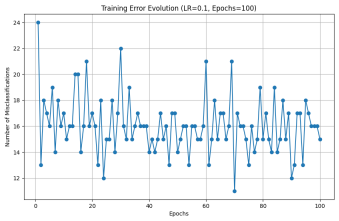
(a) LR=0.01, Epochs=100



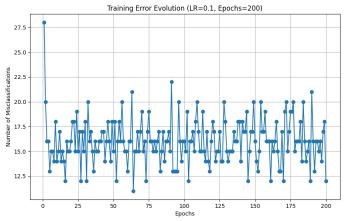
(b) LR=0.01, Epochs=200



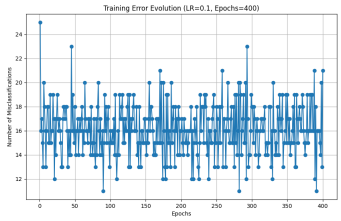
(c) LR=0.01, Epochs=400



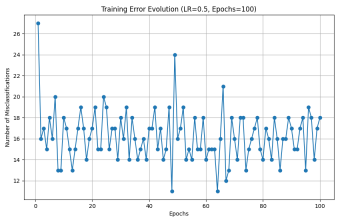
(d) LR=0.1, Epochs=100



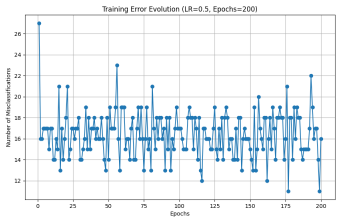
(e) LR=0.1, Epochs=200



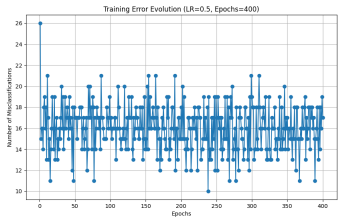
(f) LR=0.1, Epochs=400



(g) LR=0.5, Epochs=100



(h) LR=0.5, Epochs=200



(i) LR=0.5, Epochs=400

Figure 5: Training Error Evolution for different hyperparameters (Dataset 3)